Generalized Additive Model for Council District I (2023 - 2025 Sales)

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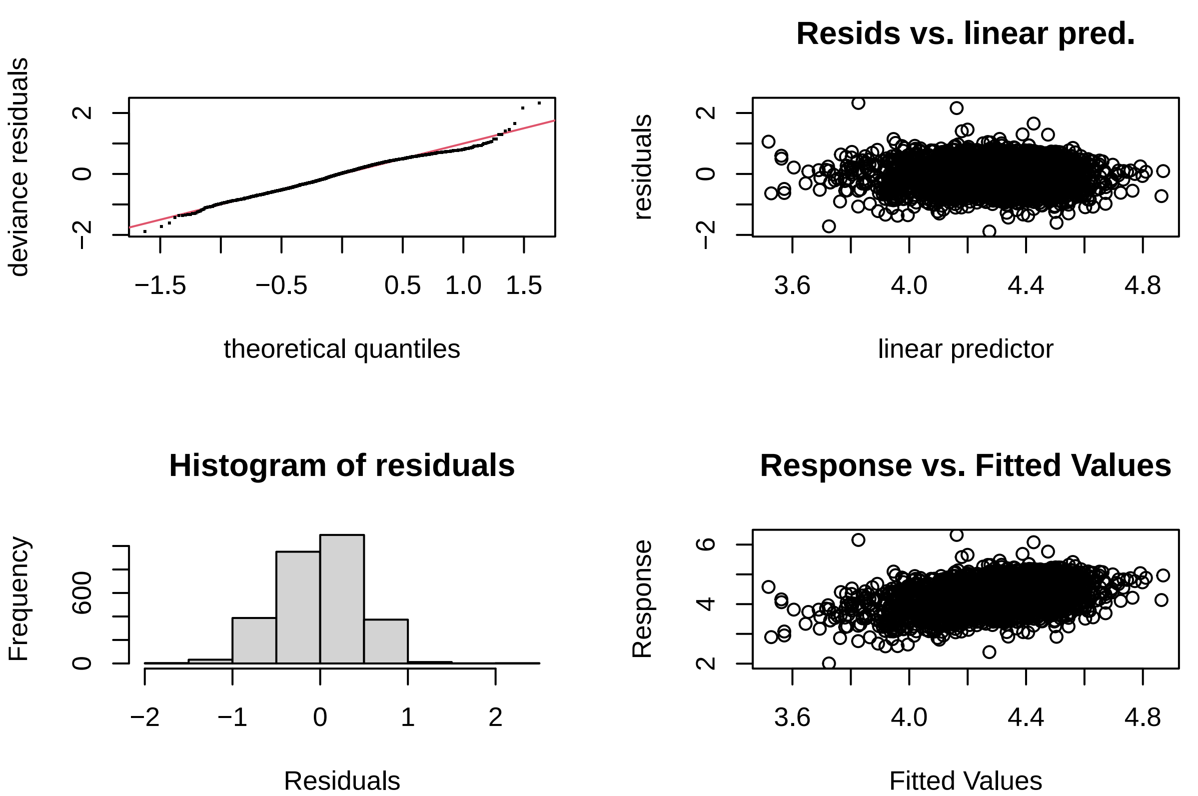
Contents

Model I

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_sppsf ~ arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          4.283953   0.033217 128.967  < 2e-16 ***
## arterial            -0.033711   0.035125  -0.960  0.33727
## single_family1story  0.027912   0.028485   0.980  0.32723
## half_duplex1story   -0.025234   0.121559  -0.208  0.83557
## half_duplex2story   -0.160581   0.190639  -0.842  0.39967
## smallest             0.071248   0.033816   2.107  0.03521 *
## small                0.037922   0.024409   1.554  0.12039
## large               -0.023294   0.027614  -0.844  0.39899
## largest              0.040733   0.055055   0.740  0.45945
## two_plus_baths       0.044208   0.074114   0.596  0.55090
## powder_room          0.008981   0.037346   0.240  0.80997
## good                 0.113675   0.063982   1.777  0.07573 .
## fair                -0.202944   0.023019  -8.816  < 2e-16 ***
## poor                -0.355458   0.062682  -5.671 1.57e-08 ***
## very_poor_unsound   -0.435543   0.138580  -3.143  0.00169 **
## qabove_avg           0.322763   0.149625   2.157  0.03108 *
## qbelow_avg          -0.028600   0.021790  -1.312  0.18946
```

```
## crawl               -0.321365   0.063838  -5.034 5.10e-07 ***
## slab                -0.279781   0.070250  -3.983 6.99e-05 ***
## fireplace            0.091012   0.022179   4.104 4.18e-05 ***
## hot_water            0.002581   0.040314   0.064  0.94895
## elec_baseboard      -0.019831   0.333176  -0.060  0.95254
## wall                 0.089394   0.095922   0.932  0.35144
## central_air          0.067002   0.023769   2.819  0.00485 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                       edf Ref.df      F p-value
## s(sresb_yearbuilt) 3.300      9  6.651  <2e-16 ***
## s(ln_garagespaces) 1.678      3 15.637  <2e-16 ***
## s(months)          4.500      9  7.580  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.133   Deviance explained = 14.3%
## -REML =   1860  Scale est. = 0.20722   n = 2855
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-7.505394e-07,1.733514e-09]
## (score 1860.027 & scale 0.2072173).
## Hessian positive definite, eigenvalue range [0.4626058,1415.506].
## Model rank =  45 / 45
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
```

```
## indicate that k is too low, especially if edf is close to k'.
##
##                       k'  edf k-index p-value
## s(sresb_yearbuilt) 9.00 3.30    0.91  <2e-16 ***
## s(ln_garagespaces) 3.00 1.68    0.90  <2e-16 ***
## s(months)          9.00 4.50    0.90  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod   prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl> <dbl>
## 1  2855     0.000812   0.000926    0.000706  1.31  46.3 -1.02
```

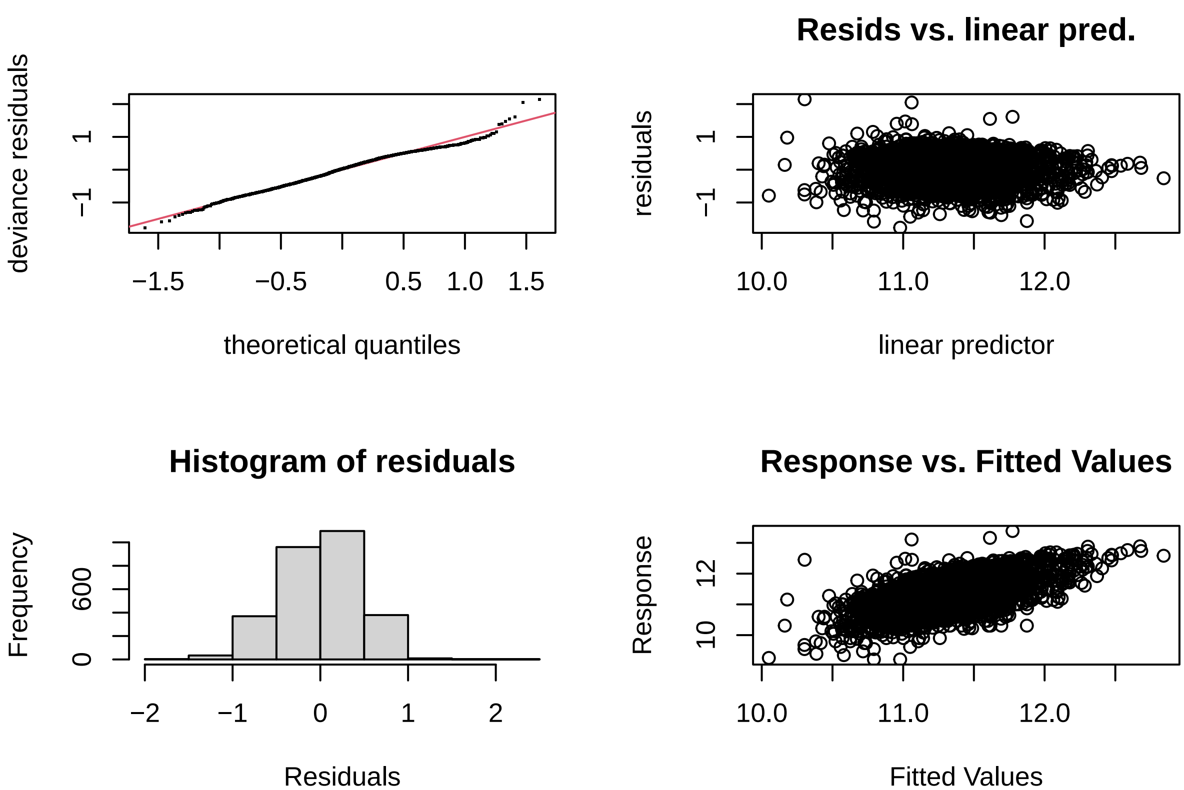
Table 1: District 1 LN_SPPSF NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2855	0.0008866	0.0010059	0.0007665	1.312344	45.77644	-0.000887

Model II

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## ln_price ~ s(ln_landratio, bs = "cs", k = 10) + s(ln_base_bldgsize_ratio,
##     bs = "cs", k = 10) + arterial + single_family1story + half_duplex1story +
##     half_duplex2story + smallest + small + large + largest +
##     two_plus_baths + powder_room + good + fair + poor + very_poor_unsound +
##     qabove_avg + qbelow_avg + s(sresb_yearbuilt, bs = "cs", k = 10) +
##     crawl + slab + s(ln_garagespaces, bs = "cs", k = 4) + fireplace +
##     hot_water + elec_baseboard + wall + central_air + s(months,
##     bs = "cs", k = 10)
##
## Parametric coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)          11.327770   0.034432 328.993  < 2e-16 ***
## arterial             -0.004416   0.034924  -0.126 0.899388
## single_family1story   0.008604   0.028359   0.303 0.761612
## half_duplex1story    -0.140227   0.123116  -1.139 0.254807
## half_duplex2story    -0.315176   0.189853  -1.660 0.097003 .
## smallest             -0.128801   0.068945  -1.868 0.061840 .
## small                -0.041046   0.035534  -1.155 0.248139
## large                 0.062287   0.039081   1.594 0.111093
## largest               0.168375   0.073718   2.284 0.022442 *
## two_plus_baths        0.183920   0.092653   1.985 0.047237 *
## powder_room           0.060767   0.042573   1.427 0.153588
## good                  0.144184   0.063397   2.274 0.023023 *
## fair                 -0.193476   0.022806  -8.484  < 2e-16 ***
## poor                 -0.325346   0.062188  -5.232 1.80e-07 ***
```

```
## very_poor_unsound   -0.370300   0.137365  -2.696 0.007065 **
## qabove_avg           0.349081   0.150175   2.324 0.020170 *
## qbelow_avg          -0.047144   0.021729  -2.170 0.030115 *
## crawl               -0.283168   0.063531  -4.457 8.63e-06 ***
## slab                -0.270503   0.069971  -3.866 0.000113 ***
## fireplace            0.114865   0.022064   5.206 2.07e-07 ***
## hot_water           -0.003831   0.039723  -0.096 0.923183
## elec_baseboard       0.057741   0.329834   0.175 0.861045
## wall                 0.061203   0.095160   0.643 0.520175
## central_air          0.067072   0.023488   2.856 0.004328 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                             edf Ref.df      F  p-value
## s(ln_landratio)           3.297      9  6.740  < 2e-16 ***
## s(ln_base_bldgsize_ratio) 1.464      9  1.256 0.000519 ***
## s(sresb_yearbuilt)        2.814      9  4.983  < 2e-16 ***
## s(ln_garagespaces)        1.592      3 10.503  < 2e-16 ***
## s(months)                 4.310      9  8.250  < 2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.357   Deviance explained = 36.5%
## -REML = 1832.6  Scale est. = 0.20256   n = 2855
```




```
##
## Method: REML   Optimizer: outer newton
## full convergence after 6 iterations.
## Gradient range [-7.329121e-05,1.301497e-05]
```

```
## (score 1832.622 & scale 0.2025618).
## Hessian positive definite, eigenvalue range [0.184497,1415.507].
## Model rank =  63 / 63
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                              k'  edf k-index p-value
## s(ln_landratio)           9.00 3.30    0.92  <2e-16 ***
## s(ln_base_bldgsize_ratio) 9.00 1.46    0.96    0.02 *
## s(sresb_yearbuilt)        9.00 2.81    0.93  <2e-16 ***
## s(ln_garagespaces)        3.00 1.59    0.91  <2e-16 ***
## s(months)                 9.00 4.31    0.91  <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
## # A tibble: 1 x 7
##       n median_ratio mean_ratio wmean_ratio   prd   cod    prb
##   <int>        <dbl>      <dbl>       <dbl> <dbl> <dbl>  <dbl>
## 1  2855        0.897       1.02       0.837  1.22  41.1 -0.698
```

Table 2: District 1 LN_PRICE (No ECFs) NLM I Assessing Coefficients

n	median_ratio	mean_ratio	wmean_ratio	PRD	COD	PRB
2855	0.9747969	1.107624	0.9090016	1.218507	40.72104	-0.7340059